

2.9 EINSTEIN COEFFICIENTS AND THEIR RELATIONS

2.9.1 Einstein Coefficients

We summarize here the three Einstein coefficients, which are the proportionality constants introduced in the above discussions.

- (i) The probability that an absorption transition occurs is given by

$$P_{12} = B_{12} \rho(\nu)$$

where B_{12} is the constant of proportionality known as the *Einstein coefficient for induced absorption*.

B_{12} is a constant characteristic of the atom and represents the properties of the energy states E_1 and E_2 .

- (ii) The probability that a spontaneous transition occurs is given by

$$(P_{21})_{\text{Spontaneous}} = A_{21}$$

where A_{21} is a constant known as the *Einstein coefficient for spontaneous emission*.

A_{21} is a constant characteristic of the atom and is known as the *radiative rate* measured in units of S^{-1} .

$(1/A_{21})$ is the lifetime of the upper state against spontaneous decay to the lower state.

- (iii) The probability that a stimulated transition occurs is given by

$$(P_{21})_{\text{stimulated}} = B_{21} \rho(\nu)$$

where B_{21} is the constant of proportionality known as the *Einstein coefficient for stimulated emission*.

B_{21} is a constant characteristic of the atom and represents the properties of the energy states E_1 and E_2 .

Note on the Einstein Coefficients.

- (a) The coefficients indicated by B are related to the Induced Transitions, *i.e.*, transitions induced by external photons.

B_{12} denotes the transition induced by a photon from lower energy level E_1 to the higher energy level E_2 .

B_{21} denotes the transition induced by a photon from higher energy level E_2 to the lower energy level E_1 .

B_{12} and B_{21} are equal when the quantum states E_1 and E_2 are single energy levels (*i.e.*, non-degenerate levels).

- (b) The coefficient indicated by A is related to the spontaneous transition, *i.e.*, transition occurred on its own without the assistance of external agent.

Since a spontaneous transition cannot take place from lower energy state E_1 to the higher energy state E_2 , we do not have the coefficient A_{12} . In other words, $A_{12} = 0$.

2.9.2 Relation between the Einstein Coefficients

The Einstein coefficients A_{21} , B_{12} and B_{21} are interrelated. To find out the relation, we assume that

- The atoms and the radiation are in thermal equilibrium.
- The radiation is identical with black body radiation and consistent with Planck's radiation law for any value of T .
- The population densities N_1 and N_2 at the lower and upper energy levels respectively are constant in time and are distributed according to Boltzmann law in the energy levels.

The above conditions require that the rate of change of atoms at the level E_2 must equal to zero. It means that the number of transitions from E_2 to E_1 must be equal to the number of transitions from E_1 to E_2 (see Fig.2.13).

Thus,

$$\left. \begin{array}{l} \text{The number of atoms absorbing} \\ \text{photons per second per unit volume} \end{array} \right\} = \left. \begin{array}{l} \text{The number of atoms emitting} \\ \text{photons per second per unit volume} \end{array} \right\}$$

$$\left. \begin{array}{l} \text{The number of atoms absorbing} \\ \text{photons per second per unit volume} \end{array} \right\} = B_{12} \rho(\nu) N_1$$

$$\left. \begin{array}{l} \text{The number of atoms emitting} \\ \text{photons per second per unit volume} \end{array} \right\} = A_{21} N_2 + B_{21} \rho(\nu) N_2$$

As the number of transitions from E_1 to E_2 must equal the number of transitions from E_2 to E_1 , we have

$$B_{12} \rho(\nu) N_1 = A_{21} N_2 + B_{21} \rho(\nu) N_2 \quad \dots(2.36)$$

$$\rho(\nu) [B_{12} N_1 - B_{21} N_2] = A_{21} N_2$$

$$\therefore \rho(\nu) = \frac{A_{21} N_2}{[B_{12} N_1 - B_{21} N_2]} \quad \dots(2.37)$$

By dividing both the numerator and denominator on the right hand side of the above equation with $B_{12} N_2$, we obtain

$$\rho(\nu) = \frac{A_{21}/B_{12}}{\left[\frac{N_1}{N_2} - \frac{B_{21}}{B_{12}} \right]} \quad \dots(2.38)$$

But $\frac{N_2}{N_1} = e^{-(E_2 - E_1)/kT}$

As $E_2 - E_1 = h\nu$, $\frac{N_2}{N_1} = e^{-h\nu/kT}$ or $\frac{N_1}{N_2} = e^{h\nu/kT}$

$$\therefore \rho(\nu) = \frac{A_{21}}{B_{12}} \left[\frac{1}{e^{h\nu/kT} - B_{21}/B_{12}} \right] \quad \dots(2.39)$$

To maintain thermal equilibrium, the system must release energy in the form of electromagnetic radiation. It is required that the radiation be identical with black body radiation and be consistent with Planck's radiation law for any value of T .

According to Planck's law

$$\rho(\nu) = \left(\frac{8\pi h\nu^3 \mu^3}{c^3} \right) \left[\frac{1}{e^{h\nu/kT} - 1} \right] \quad \dots(2.40)$$

where μ is the refractive index of the medium and c is the velocity of light in free space.

Energy density $\rho(\nu)$ given by Eq. (2.39) will be consistent with Planck's law, Eq. (2.40), only if

$$\frac{A_{21}}{B_{12}} = \frac{8\pi h\nu^3 \mu^3}{c^3} \quad \dots(2.41)$$

and $\frac{B_{21}}{B_{12}} = 1$ or $B_{12} = B_{21}$...(2.42)

Eq. (2.41) and Eq. (2.42) are known as the **Einstein relations**. The coefficients are related through

$$B_{12} = B_{21} = \frac{c^3}{8\pi h\nu^3 \mu^3} A_{21} \quad \dots(2.43)$$

- (i) Eq. (2.42) shows that the coefficients for both absorption and stimulated emission are numerically equal. The equality implies that when an atom with two energy levels is placed in the radiation field, the probability for an upward (absorption) transition is equal to the probability for a downward (stimulated) transition.
- (ii) Eq. (2.41) shows that the ratio of coefficients of spontaneous versus stimulated emission is proportional to the third power of frequency of the radiation. This is why it is difficult to achieve laser action in higher frequency ranges such as x-rays.

2.10 LIGHT AMPLIFICATION

Light amplification requires that stimulated emission occur almost exclusively. In practice, absorption and spontaneous emission always occur together with stimulated emission. The laser operation is achieved when stimulated emission exceeds in a large way the other two processes. Let us now look at the conditions under which the number of stimulated transitions can be made larger than the other two transitions.

2.10.1 Condition for Stimulated Emission to Dominate Spontaneous Emission

The ratio of Eq. (2.33) to Eq. (2.31) gives

$$\frac{\text{Stimulated transitions}}{\text{Spontaneous transitions}} = \frac{B_{21}N_2\rho(\nu)}{A_{21}N_2} = \frac{B_{21}}{A_{21}}\rho(\nu) \quad \dots(2.44)$$

Eq. (2.44) indicates that stimulated transitions will dominate the spontaneous transitions if the radiation density $\rho(\nu)$ is very large. Thus, the presence of a large number of photons in the active medium is required. However, it will lead to more absorption transitions. Hence, large photon density alone will not guarantee more stimulated emissions.

2.10.2 Requirement of States of Larger Lifetimes

Eq. (2.44) further indicates that stimulated transitions will dominate the spontaneous transitions if the value of the ratio $(B_{21})/(A_{21})$ is also large. To increase the probability of stimulated emissions, the lifetime of atoms at the excited state should be larger. In other words, it is necessary that the excited state has a longer lifetime (remember that $1/A_{21}$ represents the lifetime of the excited state).

2.10.3 Condition for Stimulated Emission to Dominate Absorption Transitions

The ratio of Eq. (2.33) to Eq. (2.29) yields

$$\frac{\text{Stimulated transition}}{\text{Absorption transition}} = \frac{B_{21}N_2\rho(\nu)}{B_{12}N_1\rho(\nu)} = \frac{N_2}{N_1} \quad \dots(2.45)$$

We used here the fact that $B_{12} = B_{21}$.

The above condition indicates that the stimulated transitions will overwhelm the absorption process if N_2 is greater than N_1 . It means that there should be more atoms present in the higher energy level than in the lower energy level for stimulated emissions to dominate over the spontaneous emissions.